

MATH 1075 - Section 8.3 - 8.4 Worksheet

10/08/2026

1 Linear, Constant, and Quadratic Function

Identify if the function is linear, constant, quadratic, or neither

1. $f(x) = x^2 + 5x + 2$

2. $f(x) = \frac{1}{2}$

3. $f(x) = \frac{3}{2}x - \frac{1}{4}$

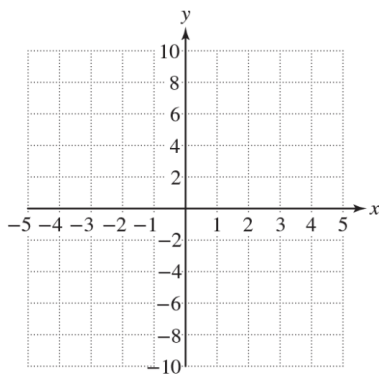
4. $f(x) = \left| \frac{2}{3}x + 6 \right|$

5. $f(x) = \sqrt{5}$

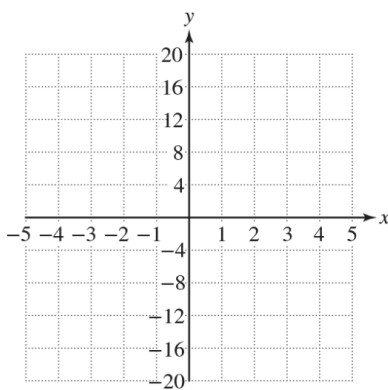
2 x -intercept, y -intercept

Find the x -intercept(s) and y -intercept of the function and graph the function.

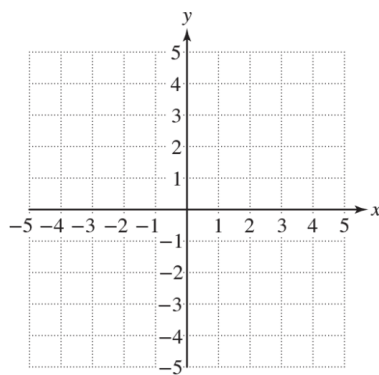
1. $f(x) = 2x + 9$



2. $f(x) = 18$



3. $f(x) = x^2 - 1$



3 Algebra and Composition of Functions

Given $f(x) = x + 4$, $g(x) = 2x^2 + 4x$, $h(x) = x^2 + 1$ find:

1. $(f + g)(x)$

2. $(g - f)(x)$

3. $(f \cdot h)(x)$

4. $\left(\frac{f}{h}\right)(x)$

5. $(f + g)(2)$

6. $(g - f)(1)$

7. $(f \cdot h)(0)$

8. $\left(\frac{f}{h}\right)(3)$

Given $f(x) = \sqrt{x+6}$ and $g(x) = x^2 + 1$, find:

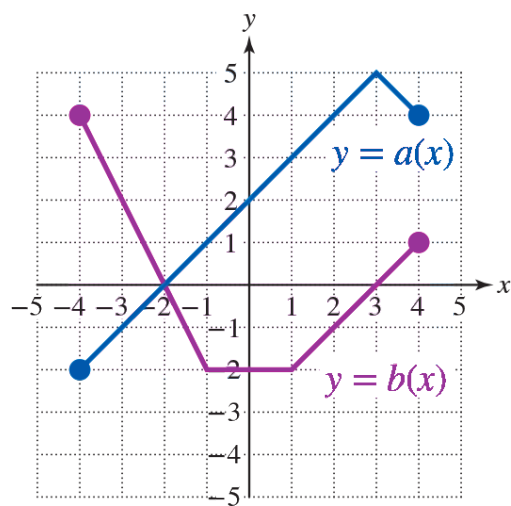
1. $(f \circ g)(x)$

2. $(g \circ f)(x)$

3. $(f \circ g)(3)$

4. $(g \circ f)(3)$

Given a graph find:



1. $a(-1)$

2. $(a + b)(1)$

3. $(a \circ b)(-2)$

4. $(b \circ a)(0)$